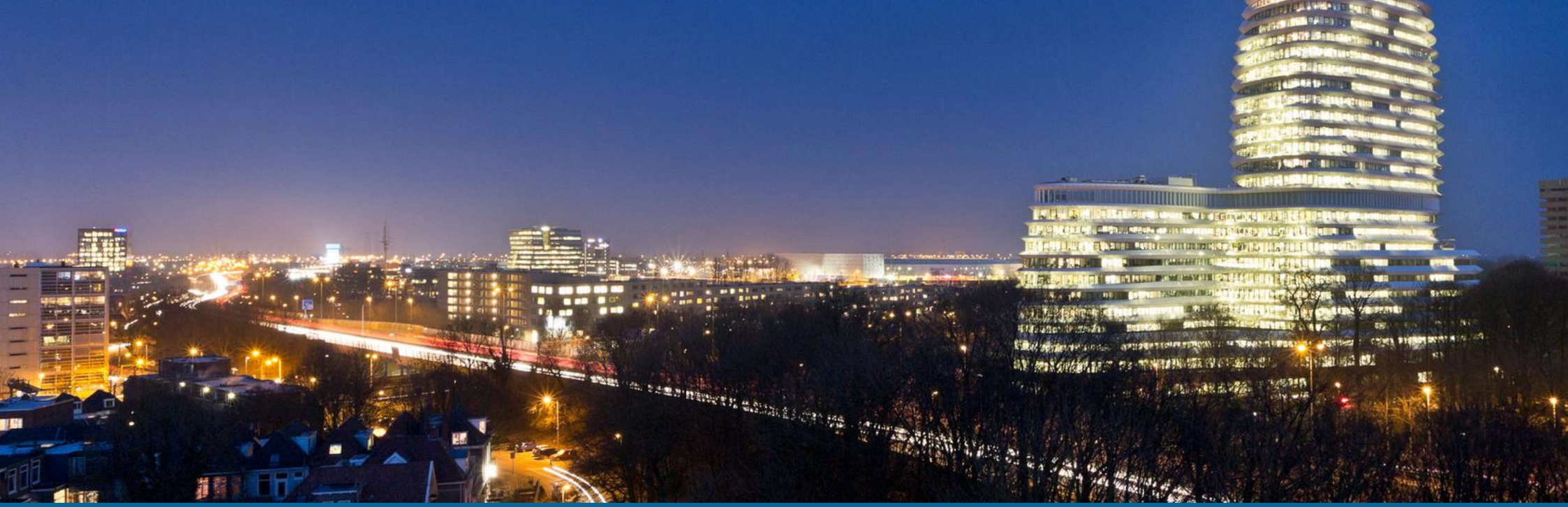

AiOps – A vision of automation and control

CD-Linux

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Agenda

- What is AiOps?
- Previous works: The belastingdienst misadventure
- A new vision: Epic CD-3324
- Pitfall! A crisis of reproducibility
- A design proposal
- Discussion

Let me take a moment…

What is AiOps: Beyond the hype

- What is real verses what is expected
  - Aggregate data to discover operational patterns of a cluster and classify them (looking for rules)
  - Know when to use unstructured learning verses supervised learning
  - Leverage change-impact analysis
  - Has found suitability in the markets with testing
    - Low-hanging fruit
- Tools verses new development
  - Quality of tools mixed and indeterminate
  - Is AI “in there” or are they just saying it is?
  - Backpropagation feedforward neural networks that can remember

This has been tried before: The belastingdienst

- Lessons learned
  - Don’t automatically trust the algorithm
  - Involve participants to determine post-mortem
  - Trust what is possible not what is theoretically possible
- Changing course
  - Determine a mathematically-sound workflow
  - Model processes in a logical manner
  - Keep functionality simple
  - Orient the process toward increasing optimization

A new vision: Epic CD-3324

Smart automation

- An executable runtime having the ability to remember changes in system variables differentiating in time can be tailored to perform complex tasks. Coupled to mechanism, it will discover the operational equilibrium of any system it is tasked within.
- If a program can reliably analyze patterns of time-series events over a window of data by paying attention to the causal patterns that are chosen for it to obser access to requisite remedies, it will apply its analysis of the best performance metric to the system it is tasked within.
- If a modal is trained to understand the relationship between causal events over a length of temporal data and the consequences of actions taken by the progra supervise it by tuning its neural network toward achieving systematic equilibrium.
- A focus on metrics and balanced behaviour in the cluster by applying an attention-focused machine-learning program that generates code and publishes (Helr OpenShift.

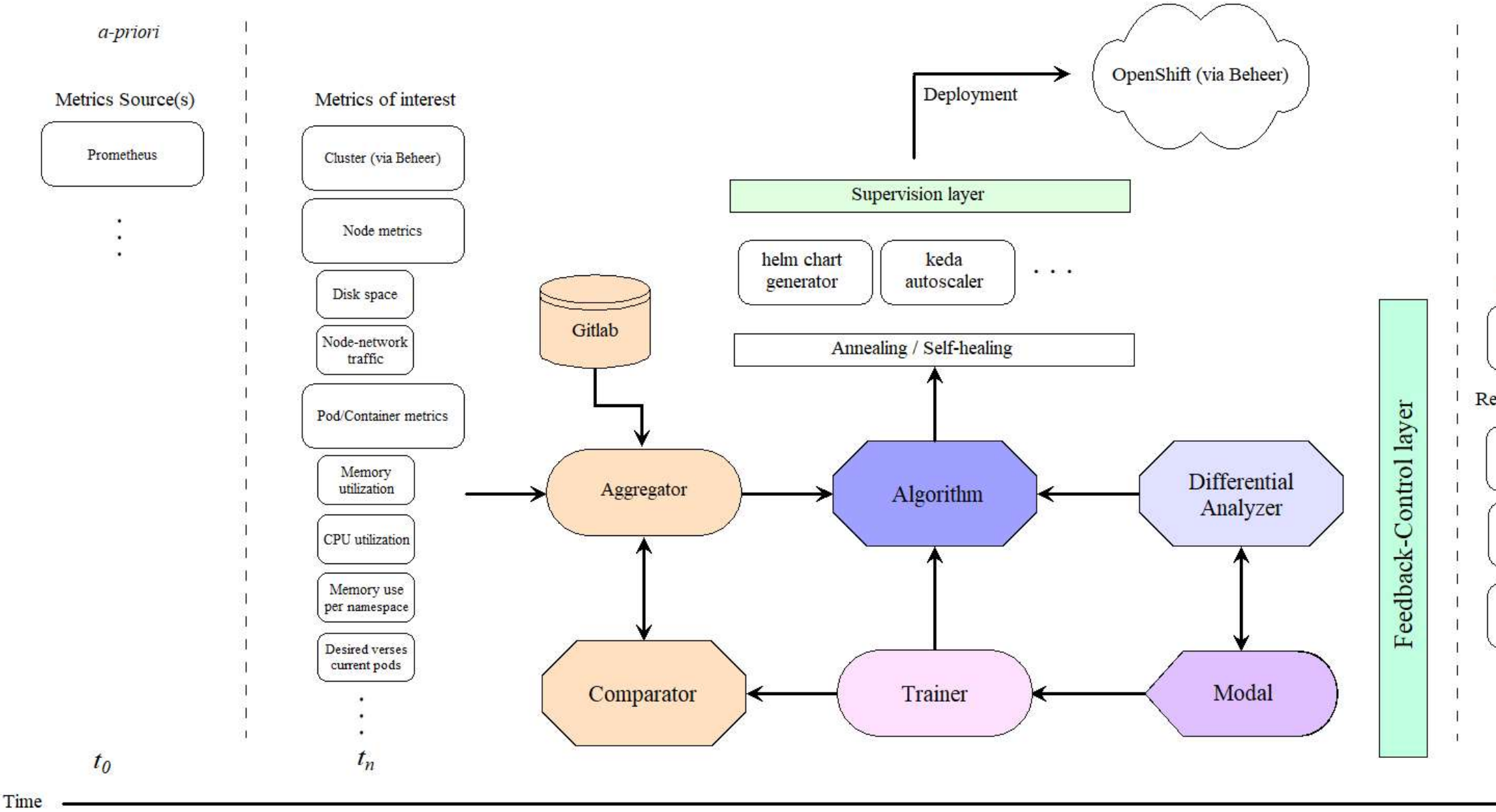
Relevant points and features

- What are the metrics the team is most concerned about keeping in check?
- Can we state we know what is the ideal metric for a given resource?
- Can we accept a solution that is arrived-at by computation?
- Training is performed as a build step.
- Output charts can be tested before they are deployed.
- Output charts can be interrupted (reviewed) as a part of the testing (confidence-building) step.
- Error/failure handling: If the agent experiences errors that it cannot solve, an option is to call a human reviewer, as notifications, generally, are und

Pitfall! A crisis of reproducibility

- Too often these methods are not done well
  - One design to suit many kinds of purposes
  - Division in machine-learning elements: Separation is key
  - Follow where the predictive mathematical solutions lead
- Collaborate
  - Don’t get intimidated by the complexity – an unavoidable feature
  - Establish fitness and create trust
  - Keep supervision tasks until (the feeling is) it is ready
  - Remember: Despite the flowery language...it is (piecewise) linear algebra...well...except for curved solu

Design proposal



Discussion points

- Is AI trustworthy?
- Can this idea be replicated to perform other kinds of tasks?
- A containerized deployment of an application?
- When is the moment we “let go” and set the algorithm free?
- Other points?